working out the odds in a craps game, it does involve a couple of steps that we must describe and then link together. Depending on your particular taste for detail, there may come a point when you just want the punch line. If this happens, feel free to jump to this page, where you'll find a summary and a discussion of conclusions stemming from Bell's discovery.

Bell and Spin

John Bell transformed the central idea of the Einstein-Podolsky-Rosen paper from philosophical speculation into a question that could be answered by concrete experimental measurement. Surprisingly, all he needed to accomplish this was to consider a situation in which there were not just two features—for instance, position and velocity—that quantum uncertainty prevents us from simultaneously determining. He showed that if there are three or more features that simultaneously come under the umbrella of uncertainty—three or more features with the property that in measuring one, you contaminate the others and hence can't determine anything about them—then there *is* an experiment to address the reality question. The simplest such example involves something known as *spin.*

Since the 1920s, physicists have known that particles spin—they execute rotational motion akin to a soccer ball's spinning around as it heads toward the goal. Quantum particle spin, however, differs from this classical image in a number of essential ways, and foremost for us are the following two points. First, particles—for example, electrons and photons— can spin only clockwise or counterclockwise at one never-changing rate about any particular axis; a particle's spin axis can change directions but its rate of spin cannot slow down or speed up. Second, quantum uncertainty applied to spin shows that just as you can't simultaneously determine the position and the velocity of a particle, so also you can't simultaneously determine the spin of a particle about more than one axis. For example, if a soccer ball is spinning about a northeast-pointing axis, its spin is shared between a northward- and an eastward-pointing axis—and by a suitable measurement, you could determine the fraction of spin about each. But if you measure an electron's spin about any randomly chosen axis, you never find a fractional amount of spin.